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Isotope selective three-step photoionization of ^{174}Yb

M. V. SURYANARAYANA* AND M. SANKARI

Bhabha Atomic Research Centre, Visakhapatnam, Andhra Pradesh, India
**Corresponding author: suryam@barc.gov.in*

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A three-step photoionization scheme has been theoretically studied for the laser isotope separation of ^{174}Yb from natural Yb using density matrix formalism. The effects of bandwidth and peak power density of lasers, Doppler broadening, and the number density of atoms on the degree of enrichment and production rate have been studied, and the optimum conditions for the efficient separation of ^{174}Yb have been derived. It has been shown that it is possible to enrich the ^{174}Yb isotope up to ∼90% using lasers with a bandwidth of 500 MHz with a production rate up to 48 mg/h. On the whole, the system required for the laser isotope separation of ^{174}Yb is much simpler than that required for the separation of any of the ^{176}Lu, ^{177}Lu, and ^{176}Yb isotopes. 1 g of the enriched isotope mixture upon irradiation for 21 days in widely available low flux reactors produces 759 doses (7.4 GBq each) of ^{175}Yb radionuclide for peptide receptor radionuclide therapy (PRRT) with 98.6% radionuclidic purity. It is envisaged that the production of the ^{175}Yb isotope through this route will be economical and widely available to the public at large as a PRRT drug for cancer therapy. © 2022 Optica Publishing Group

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1. INTRODUCTION

Radionuclides releasing β particles are increasingly being used in peptide receptor radionuclide therapy (PRRT) also known as targeted radionuclide therapy (TRT) for the treatment of neuro-endocrine tumors [1]. In PRRT, the patient receives an intravenous injection of a drug such as octreotide (DOTATOC) and octreotate (DOTATATE) that is chemically bound to a radionuclide. The β particles released by the radionuclide cause damage to the diseased cells. The mean penetration depth of β particles emitted by ^{177}Lu in soft tissue is about 670 μm. As it delivers energy to small volumes, ^{177}Lu is found to be effective in localizing cytotoxic radiation in small areas and excellent in destroying small tumors as well as metastatic lesions, which are typically about <3 mm diameter.

The choice of a radionuclide for therapeutic applications is determined by its characteristics such as type and energy of radiation, half-life, radionuclidic purity, specific activity, ease of production, cost, and availability of the enriched precursor isotope. Several radionuclides for PRRT are produced by irradiation of precursor isotopes in nuclear reactors [2]. As per the International Atomic Energy Agency (IAEA) data of 2022, there are about 60 low flux reactors (neutron flux $<10^{14}$ neutrons/cm^2 − s), 44 medium flux reactors (neutron flux $10^{14} - 10^{15}$ neutrons/cm^2 − s), and 7 high flux reactors (neutron flux $>10^{15}$ neutrons/cm^2 − s) worldwide currently in operation. Therefore, the two primary factors that are detrimental to the availability of an affordable and widely available PRRT drug to the public at large are the easy and efficient enrichment process of the precursor isotope and the production of the radionuclide in low to medium flux reactors.

The ^{177}Lu isotope can be produced in a nuclear reactor by the irradiation of its enriched parent isotope, either the ^{176}Lu or ^{176}Yb isotope. A detailed review of the methods for the production of ^{177}Lu can be found in [3]. Production of ^{177}Lu through irradiation of enriched ^{176}Yb in low to medium flux reactors is rather an inefficient process due to its low thermal neutron absorption cross section of ^{176}Yb ($\sigma = 2.85$ barns) [3]. Further, the extremely cumbersome chemical separation process [4] for the separation of micro amounts of ^{177}Lu produced during irradiation from the large excess of unirradiated ^{176}Yb isotope makes this process rather unattractive.

On the other hand, production of ^{177}Lu through irradiation of enriched ^{176}Lu is far more efficient due to its high thermal neutron absorption cross section of ^{176}Lu ($\sigma = 2090$ barns). However, this route is limited by its inability to produce highly enriched (>95%) ^{176}Lu precursor isotope through the atomic vapor laser isotope separation (AVLIS) process. This is due to the highly complex atomic hyperfine structure and small isotope shifts of the constituent Lu isotopes which results in the production of low specific activity ^{177}Lu. In case of ^{176}Lu, despite utilization of narrowband lasers (bandwidth 100 MHz)

for the laser isotope separation process, the degree of enrichment was reported to be 68.9% [5,6]. When 68.9% enriched ^{176}Lu is irradiated in a low flux nuclear reactor (neutron flux 3×10^{13} neutrons/cm^2 − s) for 21 days, it produces ^{177}Lu with an isotopic purity of 3% with specific activity ∼130 TBq/g while the no carrier added specific activity is 4104 TBq/g. Thus, despite the high thermal neutron absorption cross section of ^{176}Lu, low degree of enrichment of ^{176}Lu and irradiation in the low flux reactors inhibit production of high specific activity ^{177}Lu. Alternatively, it has also been proposed to produce high specific activity ^{177}Lu by irradiation of natural Lu in a nuclear reactor followed by laser isotope separation of the target isotope [7–11]. However, this involves handling of radioactive isotopes during the laser separation process. On the whole, the enrichment process of the radionuclides or its precursors of Lu and Yb, viz., ^{176}Lu, ^{177}Lu, and ^{176}Yb require narrowband lasers (bandwidth ∼100 MHz) for the isotope separation. For all the reasons mentioned above, the search for a suitable radionuclide is currently in progress, which will enable easy and economical production [12].

The ^{175}Yb isotope has been investigated as an attractive isotope for PRRT [13,14] as it has a convenient half-life of 4.1615 days [15] and is comparable to the 6.65 day half-life of ^{177}Lu. Further, the decay characteristics of these two isotopes such as energies of the β particles and γ rays are very similar (Table 1). ^{175}Yb can be produced by neutron irradiation of ^{174}Yb in a nuclear reactor through the nuclear reaction ^{174}Yb $(n, \gamma)^{175}$Yb. Further, it may be noted that the thermal neutron absorption cross section and natural abundance of ^{174}Yb (63.2 b, 31.83%) are considerably higher than ^{176}Yb (2.85 b, 12.76%). As a result of these characteristics, it is possible to convert ^{174}Yb into ^{175}Yb through neutron irradiation in low flux reactors (Fig. 1). Further, due to higher initial abundance of ^{174}Yb, it is premised that the enrichment process may be easier in comparison to the enrichment of ^{176}Yb.

In the present work, we have examined the possibility of using AVLIS for the production of highly enriched ^{174}Yb. We have adopted density matrix formalism for deriving the system parameters for the optimization of the AVLIS process. We have also theoretically evaluated the suitability of the produced ^{175}Yb from enriched ^{174}Yb for PRRT applications.

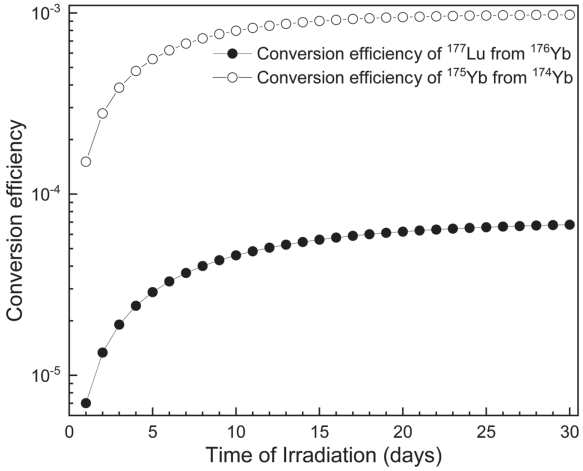


Fig. 1. Conversion efficiency of 100% enriched ^{174}Yb or ^{176}Yb precursor isotope to daughter radionuclide ^{175}Yb or ^{177}Lu, respectively, with irradiation time. The neutron flux of the reactors is 3×10^{13} neutrons/cm^2/s. Cooling time after irradiation is 0 days.

2. AVLIS OF YTTERBIUM

Yb has seven naturally occurring isotopes. The thermal neutron absorption cross section and the natural abundance of the isotopes are ^{168}Yb (2300 b, 0.13%), ^{170}Yb (9.9 b, 3.04%), ^{171}Yb (58.3 b, 14.28%), ^{172}Yb (1.30 b, 21.83%), ^{173}Yb (15.5 b, 16.23%), ^{174}Yb (63.2 b, 31.83%), and ^{176}Yb (2.85 b, 12.76%). Yb has a ground state configuration of $4f^{14}6s^2$ ($^1S_0$) and an ionization energy of 50443.2 cm^{-1} (6.254 eV). The high repetition rate nanosecond tunable excitation lasers used in the laser isotope separation process have typical photon energies of ∼ 2 eV, therefore, it is necessary to adopt a three-step photoionization process for the ionization of Yb. Based on the energy levels of Yb, oscillator strengths of the connecting transitions, and availability of the suitable lasers for accessing the transitions, the following photoionization scheme is considered to be the best available option [16–23]:

$$6s^2\,{}^1S_0(0.0\ \text{cm}^{-1}) \xrightarrow{555.8023\ \text{nm}} 6s6p\,{}^3P_1^{\circ}\ (17992.007\ \text{cm}^{-1})$$

$$\xrightarrow{581.2273\ \text{nm}} 4f^{14}6s^2\left(\frac{7}{2},\frac{3}{2}\right)J=2\ (35196.98\ \text{cm}^{-1})$$

$$\xrightarrow{582.88\ \text{nm}}\ 52353\ \text{cm}^{-1}\ \text{Autoionization State} \rightarrow \text{Yb}^+.$$

Laser–atom interactions in a multi-step laser photoionization process can be accurately described through density matrix formalism [24]. A generalized three-step photoionization scheme is shown in Fig. 2. The atoms initially present in the $|1\rangle$ ground fine structure level are excited by the first excitation laser having energy $\hbar\omega_1$ to the first excited level $|2\rangle$. The atoms from the first excited level are further excited to the second excited level $|3\rangle$ using the second excitation laser having energy $\hbar\omega_2$, which are further incoherently excited to the autoionization level by absorbing photons from a third laser and are finally ionized. The atoms in a resonant $|n\rangle$ upper level may decay to the resonant lower level $|m\rangle$ at a rate denoted as Γ_{nm}. The atoms in the resonant level $|m\rangle$ also decay to trapped level at a rate denoted as γ_m and lost from the excitation process. The population dynamics in a three-step ladder excitation of an atom are described by the coupled density matrix equations published in [6] and are also available in Supplement 1.

In the present work, ionization is considered to be an incoherent process, which is induced by the ionizing laser at a rate $\gamma_i = \sigma\Phi$, where σ is the photoionization cross section and ϕ is the flux of the ionization laser.

The effect of the laser bandwidth and its line shape are included in the terms [25,26],

$$2 \cdot \gamma_i \frac{\beta_i^2}{\Delta_i^2 + \beta_i^2}, \quad i = 1, 2. \tag{1}$$

According to the phase diffusion model, the laser spectrum is Lorentzian near the center with a full width at half-maximum (FWHM) γ_i and has a cutoff around β_i. For the detunings $\Delta_i \ll \beta_i$ these terms reduce γ_i; while $\Delta_i \gg \beta_i$, the laser appears to be monochromatic.

Table 1. Table of Nuclear Properties of ^{175}Yb and ^{177}Lu Isotopes and Their Precursors

Parent Isotope	Natural Abundance (%)	Nuclear Reaction	Thermal Neutron Absorption Cross Section (Barns)	Half-Life (Days)	Decay Mode	Energy
^{176}Lu	2.59	$^{176}\mathrm{Lu}(n,\gamma)^{177}\mathrm{Lu}$	2090	6.65	β^-	497 keV (78.6 %), 384 keV (9.1 %) and 176 keV (12.2 %)
^{176}Yb	12.76	$^{176}\mathrm{Yb}(n,\gamma)^{177}\mathrm{Yb}\xrightarrow{T_{1/2}=1.911\,h}{}^{177}\mathrm{Lu}$	2.85		γ	208 keV (11 %), 113 keV (6.6 %),
^{174}Yb	31.83	$^{174}\mathrm{Yb}(n,\gamma)^{175}\mathrm{Yb}$	63.2	4.1615	β^-	470 keV (86.5 %), 356 keV (3.3 %) and 74 keV (10.2 %)
					γ	396 keV (6.4 %), 282 keV (3.0 %) and 114 keV (1.9 %)

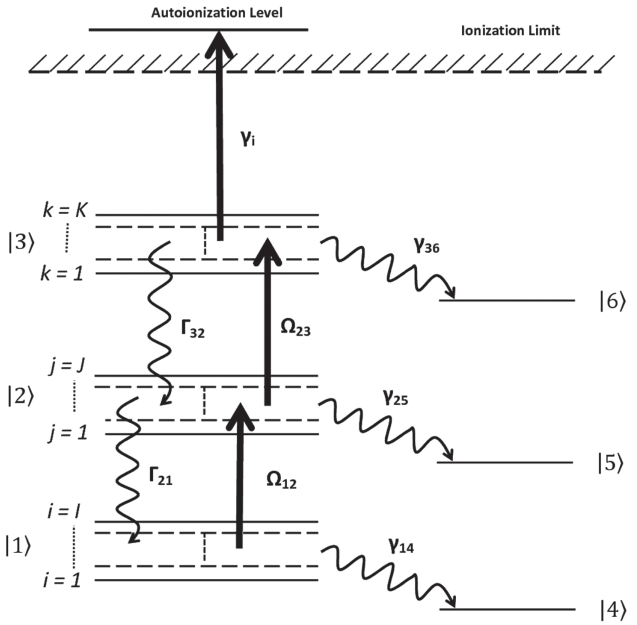


Fig. 2. Schematic diagram of the three-step photoionization of Yb (not to scale). Ω is the Rabi frequency, γ_i is the ionization rate, Γ is the radiative decay rate to the lower level, and γ is the radiative decay rate to the trapped level.

For a typical atomization temperature of 600°C for Yb, the most probable velocity of ^{174}Yb is ∼287.1 m/s. For the entire calculations in this work, the pulse width of the laser is considered to have a Gaussian pulse width of 30 ns and no delay between the pulses, and all the lasers are considered to be co-propagating. Initially, the population of the ground state is set to one. The coupled density matrix equations are then integrated using the standard numerical integration method for the set conditions for the entire duration of the laser–atom interaction. At the end of the laser interaction, the population of the ion state corresponds to the ionization efficiency of the photoionization process. For the inclusion of Doppler broadening of the atomic ensemble in the calculations, divergence angle of the atomic ensemble is segmented into a minimum of 30 angular groups; and each angular group is segmented into a minimum of 30 velocity groups.

3. RESULTS AND DISCUSSION

First, variation in the ionization efficiency of the Yb isotopes with a change in the frequency of both the excitation lasers has been calculated. Since this calculation requires enormous computational time, these calculations have been restricted to Doppler free conditions, i.e., ignoring the angular and velocity distributions of the atomic vapor. The resultant ionization efficiency plot is shown in Fig. 3. The resonance frequency positions of the stepwise transitions have been identified with serial numbers listed in Table 2. The horizontal and vertical ridges of each isotope transition correspond to the excitation transition of the first and second step, respectively. The diagonal ridges correspond to the coherent two-photon ionization wherein the sum frequency of the photons is equal to the two-step excitation process. It can also be observed from Fig. 3 that the coherent two-photon ionization lines of the ^{173}Yb isotope hyperfine pathways marked with numbers 2 (5/2–7/2–7/2 hyperfine pathway), 9 (5/2–5/2–3/2 hyperfine pathway), 10 (5/2–5/2–5/2 hyperfine pathway), and 11 (5/2–5/2–7/2 hyperfine pathway) pass through (or adjacent to) the resonance frequency position of ^{174}Yb. Though these frequency positions of ^{173}Yb hyperfine pathways are a few GHz away from the resonance position of ^{174}Yb, this non-target isotope could still be ionized if one employs broadband lasers, high power lasers, or both. As a result, the degree of enrichment of ^{174}Yb could be affected significantly. Therefore, one needs to optimize the system parameters to obtain the highest possible enrichment with highest ionization efficiency possible. The effects of various system parameters on the degree of enrichment are discussed in the following sections.

A. Power of the Excitation and Ionization Lasers

When the peak power density of the excitation and ionization lasers is set to 10 W/cm^2, the ionization efficiency of ^{174}Yb is found to be 1.0×10^{-4} (Fig. 3). This ionization efficiency is not adequate for the large scale production of enriched ^{174}Yb isotope. Therefore, one needs to increase the power of the excitation and ionization lasers. When the peak power density of the excitation laser is increased beyond the saturation power density, the saturation broadening of the lines occurs.

The saturation broadening (Γ_{sat}) can be calculated using the following expression:

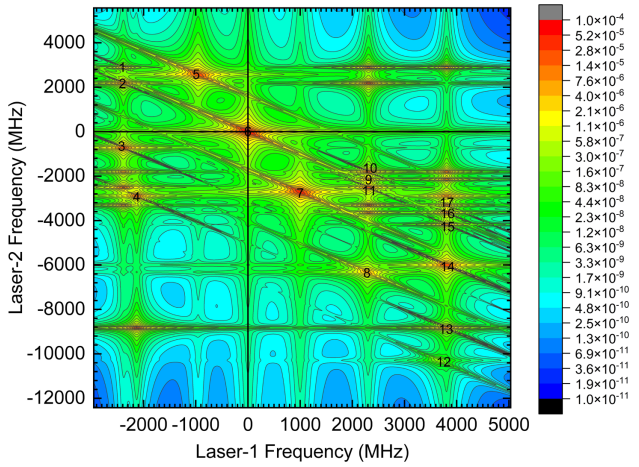


Fig. 3. Two-dimensional contour of the ionization efficiency of Yb isotopes normalized to the natural abundance under Doppler free condition. Peak power densities of all lasers are set to 10 W/cm^2, and the bandwidth of the excitation lasers is set to 10 MHz. Resonance frequency positions of the isotopes (or hyperfine pathways in case of odd isotopes) are marked with numbers as per Table 2.

$$\Gamma_{\text{sat}} = \Gamma_0\sqrt{1+\frac{I_{\text{laser}}}{I_{\text{sat}}}}, \tag{2}$$

where Γ_0 is the natural broadening width, I_{laser} is the peak power density of the laser (W/cm^2), and I_{sat} is the saturation power density (W/cm^2). The radiative lifetimes of the upper levels of the first and second excitation transitions are 840 ns and 1120 ns [16], and their corresponding natural broadening widths are 1.19 MHz and 0.89 MHz, respectively. The respective saturation power densities are 0.43 mW/cm^2 and 0.28 mW/cm^2. When the excitation laser peak power densities are set to 10 W/cm^2, the corresponding saturation broadening values are 182 MHz and 534 MHz, respectively. Since the saturation broadening is much smaller than the frequency separations of the interfering transitions of other isotopes (viz., S. No. 2, 9, 10, 11 of Table 2), isotope selective photoionization of ^{174}Yb is possible. Under such conditions, the laser peak power density of the excitation lasers can be increased for increasing the ionization efficiency.

When the peak power density of the excitation lasers is increased, the ionization efficiency of the resonant isotope increases; however, at the same time, ionization efficiency of the non-resonant isotopes also increases due to the saturation broadening of the lines. As a result, isotope selectivity decreases. Therefore, high ionization efficiency can be achieved only at the cost of isotope selectivity. At certain peak power densities, high ionization efficiency can be achieved without significant loss in the isotope selectivity.

In order to find such optimum peak power density values, a series of ionization efficiency calculations has been carried out varying peak power densities of the excitation and ionization lasers for various bandwidths of the excitation lasers under Doppler free conditions. From the computed ionization efficiencies of each isotope, the degree of enrichment has been calculated using the following expression:

$$\text{Degree of enrichment (\%) of } ^{174}\text{Yb} = \left\{\frac{\eta_{174} * A_{174}}{\sum_i^{\text{all}} \eta_i * A_i}\right\} * 100, \tag{3}$$

where η corresponds to the ionization efficiency of the isotope, A is the fractional abundance, and i is the isotope mass index.

The obtained optimum peak power density values of the lasers required to achieve a degree of enrichment of ~95% have been tabulated along with the ionization efficiency (Table 3).

Table 2. Table of the Frequency Positions of Yb Isotopes Relative to ^{174}Yb in a Three-Step Photoionization Scheme[a]

S No	Isotope	J1 (Even Isotope)/F1 (Odd Isotope)	J2 (Even Isotope)/F2 (Odd Isotope)	J3 (Even Isotope)/F3 (Odd Isotope)	Frequency Position for the First Excitation Transition (Relative to ^{174}Yb) (MHz)	Relative Intensity	Frequency Position for the Second Excitation Transition (Relative to ^{174}Yb) (MHz)	Relative Intensity
1	^{173}Yb	2.5	3.5	2.5	−2387	44.4	2888	1.6
2	^{173}Yb	2.5	3.5	3.5	−2387	44.4	2186	9.5
3	^{173}Yb	2.5	3.5	4.5	−2387	44.4	−723	33.3
4	^{171}Yb	0.5	0.5	1.5	−2132	33.3	−2891	33.3
5	^{176}Yb	0	1	2	−955	100.0	2578	100.0
6	^{174}Yb	0	1	2	0	100.0	0	100.0
7	^{172}Yb	0	1	2	1000	100.0	−2711	100.0
8	^{170}Yb	0	1	2	2287	100.0	−6318	100.0
9	^{173}Yb	2.5	2.5	1.5	2310	33.3	−2143	4.0
10	^{173}Yb	2.5	2.5	2.5	2310	33.3	−1808	12.2
11	^{173}Yb	2.5	2.5	3.5	2310	33.3	−2510	17.1
12	^{168}Yb	0	1	2	3655	100.0	−10293	100.0
13	^{171}Yb	0.5	1.5	1.5	3805	66.7	−8828	6.7
14	^{171}Yb	0.5	1.5	2.5	3805	66.7	−5996	60.0
15	^{173}Yb	2.5	1.5	0.5	3807	22.2	−4176	6.7
16	^{173}Yb	2.5	1.5	1.5	3807	22.2	−3640	9.3
17	^{173}Yb	2.5	1.5	2.5	3807	22.2	−3306	6.2

[a]The isotope shift and hyperfine structure data taken from [16].

Table 3. Table of the Degree of Enrichment of ^{174}Yb for Various Bandwidths of the Excitation Lasers under Doppler Free Condition

Bandwidth (MHz) of the Excitation Lasers	Optimum Peak Power Density (W/cm^2)			$\eta_{^{174}Yb}$	Degree of Enrichment of ^{174}Yb (%)
	555.8023 nm	581.2273 nm	582.88 nm		
50	900	3300	50,000	0.644	95.0
100	900	3100	50,000	0.641	95.0
250	700	2000	50,000	0.632	95.0
500	300	600	50,000	0.595	95.4
750	500	100	50,000	0.459	95.1
1000	100	100	50,000	0.384	94.1

Since these calculations disregard the angular and velocity distribution of the atomic ensemble, the ionization efficiency and degree of enrichment represent the upper limits. The effect of Doppler broadening on the degree of enrichment has been dealt with in the subsequent sections.

When the bandwidth of the excitation lasers is small (<500 MHz), high peak power density of the lasers can be used without the loss in the degree of enrichment. This is because the resultant broadening of line shape due to saturation broadening and laser bandwidth does not cause loss in the isotopic selectivity. As the laser bandwidth is increased, the resultant broadening also increases; therefore, in order to achieve the same degree of enrichment (∼95%), the peak power density of the excitation lasers needs to be lowered. This causes loss in the ionization efficiency of ^{174}Yb. This explains the trend in the optimum peak power densities of lasers and ionization efficiency in Table 3.

Further, the degree of enrichment falls with an increase in the bandwidth of the excitation lasers. In case of ^{174}Yb, this fall in the degree of enrichment (from ∼ 95% to 94%) with an increase in the bandwidth of the excitation lasers from 50 MHz to 1000 MHz is not significant. This is in contrast to the enrichment of ^{176}Yb, wherein a significant fall in the degree of enrichment (∼97% to 66%) has been reported with an increase in the bandwidth from 50 MHz to 500 MHz [23]. This can be primarily attributed to the higher natural abundance of ^{174}Yb (31.83%) compared with ^{176}Yb (12.76%). This can be further explained in the following.

Enrichment factor for ^{174}Yb can defined as

$$\text{Enrichment factor } \alpha = \frac{\{E_{174}/(100 - E_{174})\}}{\{N_{174}/(100 - N_{174})\}}, \tag{4}$$

where E_{174} and N_{174} are the percentage abundance of the ^{174}Yb isotope in the enriched and natural isotope mixtures.

In case of ^{174}Yb, in order to enrich the ^{174}Yb isotope from natural Yb to 95%, the enrichment factor of 41 should be obtained, while in case of ^{176}Yb for the same degree of enrichment, the requirement of the enrichment factor is 130. Thus, ^{174}Yb enrichment poses fewer challenging conditions compared with ^{176}Yb.

Though degree of enrichment is largely invariant with an increase in the bandwidth of the excitation lasers under optimum conditions, the ionization efficiency starts falling when the bandwidth of the excitation laser is >500 MHz (Table 3). Therefore, a laser bandwidth of 500 MHz can be considered optimum for the enrichment of ^{174}Yb.

B. Doppler Broadening

In the laser isotope separation process, an atomic ensemble will consist of angular and velocity distribution. The flux distribution of the atomic ensemble is given by the following expression [27]:

$$\phi(v) = 2\left(\frac{v^3}{v_{mp}^4}\right) \cdot e^{-\left(\frac{v^2}{v_{mp}^2}\right)} dv, \tag{5}$$

where v_{mp} is the most probable velocity. At $4 \times v_{mp}$, the relative flux drops to the value of $\sim 10^{-7}$ of the maximum.

The most probable velocity of the atoms can be calculated using the following equation:

$$v_{mp} = \sqrt{\frac{2kT}{m}}, \tag{6}$$

where k is the Boltzmann's constant ($1.38064852 \times 10^{-23}$ m^2 kg s^{-2} K^{-1}), T is the temperature (^{0}K), and m is mass of the atom (kg).

As mentioned earlier, for a typical atomization temperature of 600°C for Yb, the most probable velocity of ^{174}Yb is ∼287.1 m/s. The velocity component along the laser axis is calculated by the following equation:

$$v = v_{mp} \times \text{Sin}(\theta/2)\,, \tag{7}$$

where θ is the full angle divergence.

As a result of the velocity component along the laser axis, the resonance under Doppler shifts. The Doppler shift of each velocity group is calculated using the following expression:

$$f = \frac{v}{c} f_0, \tag{8}$$

where f is the Doppler shift, v is the velocity of atom along the laser axis, c is the velocity of light, and f_0 is the resonance frequency of the transition.

For the inclusion of Doppler broadening of the atomic ensemble in the calculations, the divergence angle is segmented into a minimum of 30 angular groups, and each angular group is segmented into a minimum of 30 velocity groups.

When an atomic source is operated at the temperature of 600°C, for the 555.8023 nm transition, the Doppler broadening of the unrestricted atoms will be 866 MHz. When a collimator is introduced along the perpendicular plane of the atomic beam propagation axis, all the velocity groups falling within the divergence limit of the collimator pass though and

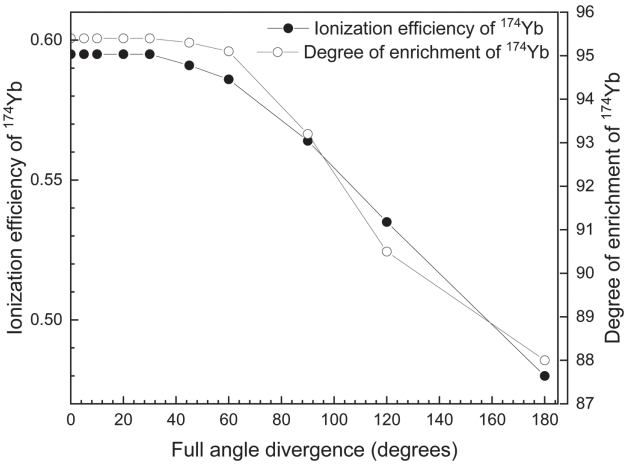


Fig. 4. Dependence of ionization efficiency and degree of enrichment of ^{174}Yb on the full angle divergence of the atomic ensemble. Bandwidth of the excitation laser is set to 500 MHz. Peak power densities of the first, second, and third excitation lasers are 300, 600, and 50,000 W/cm^2.

enter the laser–atom interaction region while remaining atoms will be prevented to enter this region. Thus, the reduction in velocity induced shifts (Doppler shifts) causes reduction in the Doppler broadening. When a collimator with full angle divergence of 10° is introduced, the Doppler broadening is reduced to 288 MHz. An increase in the full angle divergence to 30°, 45°, and 60° increases the Doppler broadening to 428 MHz, 634 MHz, and 830 MHz, respectively.

Calculations of the ionization efficiency and the degree of enrichment have been carried out varying the full angle divergence of the atomic ensemble from 0° (corresponds to Doppler free condition) up to 180°. The results are plotted in Fig. 4. It can be seen from the graph that both ionization efficiency and the degree of enrichment of the ^{174}Yb isotope largely remain largely unaffected up to the full angle divergence value of 60°. When the Doppler broadening of the atomic ensemble is more than the bandwidth of the excitation lasers, the angular and velocity groups that have higher Doppler shifts will not be excited. As a result, ionization efficiency of the target isotope is reduced. For the same reasons, the angular and velocity groups that have higher Doppler shifts of non-target isotopes could fall into the resonance. As a result, the degree of enrichment is lost. From Fig. 4, a full angle divergence of 60° can be considered as the limit for the optimum separation of ^{174}Yb.

C. Bandwidth of Excitation Lasers and Charge Exchange Collisions

After obtaining the values of optimum peak power densities of the lasers and angular divergence limit for the laser isotope separation of the ^{174}Yb isotope, the effect of the bandwidth of the excitation lasers was studied. When the bandwidth of the excitation lasers is much larger than the spectral width of the isotope (which is a result of confluence of Doppler broadening and saturation broadening), the photon economy is lost, as many of the photon frequencies will lay away from the resonance. On the other hand, when the bandwidth of the excitation lasers is much smaller than the spectral width of the isotope, the throughput is lost, as Doppler shifted resonances of the larger velocity groups will lay away from the resonance. Therefore, in order to achieve optimum efficiency for the laser isotope separation process in terms of photon economy, throughput, and degree of enrichment, it is necessary to employ lasers with appropriate bandwidth.

Further, apart from the bandwidth and peak power density of the lasers, Doppler broadening, and saturation broadening of the atomic ensemble, charge exchange collisions play a vital role in determining the achievable degree of enrichment and production rates in a laser isotope separation process. The photoions that are formed during the laser–atom interaction in a multi-step photoionization process are extracted by applying the orthogonal electric field. These photoions are likely to undergo charge exchange collisions with the unionized atomic vapor during the extraction. The probability of such a charge exchange can be calculated using the equation given below [19]:

$$p = 1 - e^{-\sigma d N}, \tag{9}$$

where σ is the resonant charge exchange cross section (cm^2), d is the distance traversed by the photoions prior to collection at the collector (cm), and N is the number density of the atoms (atoms/cm^3).

The resonant charge exchange cross section of Yb corresponding to atomic velocities at the temperature of 600°C is 2×10^{-14} cm^2 [28]. For a number density of 1×10^{12} atoms/cm^3 and a 50 mm extraction length, the probability of charge exchange collisions is calculated to be 9.5%. When the number density of the atoms is increased further by 1 order of magnitude to 1×10^{13} atoms/cm^3, the probability of charge exchange increases to 63%. At this number density, the degree of enrichment is severely degraded. Therefore, it is necessary to limit the number densities to $<1 \times 10^{13}$ atoms/cm^3 for obtaining meaningful enrichment of ^{174}Yb.

A series of computations has been carried out varying the bandwidth of both the excitation lasers wherein the degree of enrichment has been calculated for various number densities from $1 \times 10^{11} - 5 \times 10^{12}$. The results have been plotted in Fig. 5 (see also Fig. 6) for four important isotopes of the isotope mixture. Among the natural isotopes of Yb, the content of ^{168}Yb, ^{173}Yb, ^{174}Yb, and ^{176}Yb determine the usefulness of the enriched isotope mixture for PRRT applications after irradiation for the following reasons. First, the ^{168}Yb isotope, which has very high thermal neutron absorption cross section (2300 barns), produces long-lived ^{169}Yb ($T_{1/2} = 32.02$ days) radioisotope upon irradiation in a nuclear reactor. Second, the ^{173}Yb isotope, which has moderate thermal neutron absorption cross section (15.5 barns), upon irradiation in a nuclear reactor, produces ^{174}Yb, which is a precursor to ^{175}Yb medical isotopes. Third, the ^{174}Yb isotope, is the degree of enrichment of which will determine the specific activity of the ^{175}Yb medical isotope produced after irradiation. Fourth, the ^{176}Yb isotope, upon irradiation produces the ^{177}Lu radioisotope having a half-life of 6.65 days.

The other remaining natural isotopes ^{170}Yb, ^{171}Yb, and ^{172}Yb do not produce any radionuclide upon irradiation in the nuclear reactor. Further, their total abundance in the enriched

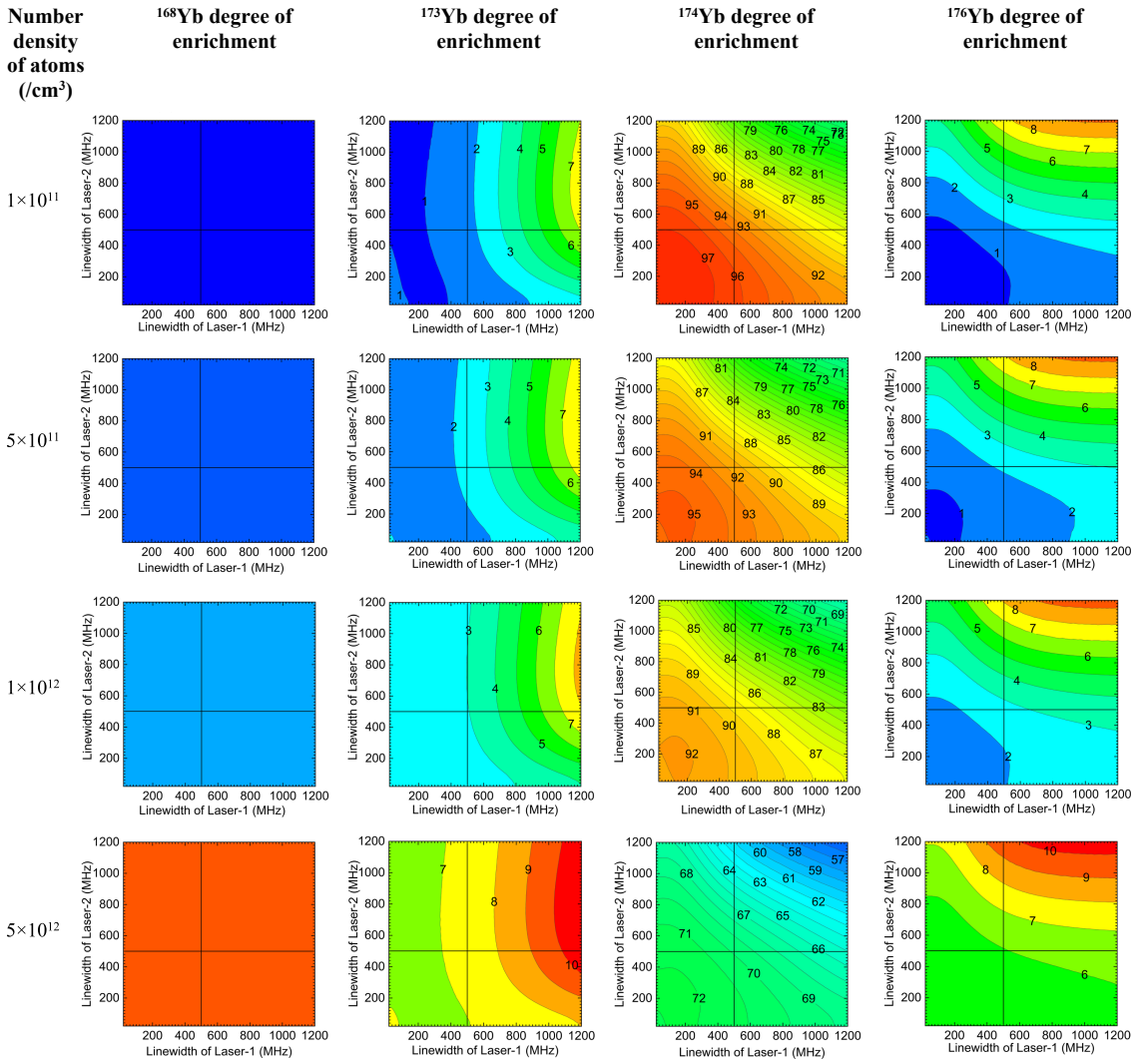


Fig. 5. Degree of enrichment of Yb isotopes with variation in the bandwidth of excitation lasers for different atomic number densities of the atomic ensemble. Full angle divergence of the atomic beam is 60°. Peak power densities of the first, second, and third excitation lasers are 300, 600, and 50,000 W/cm^2.

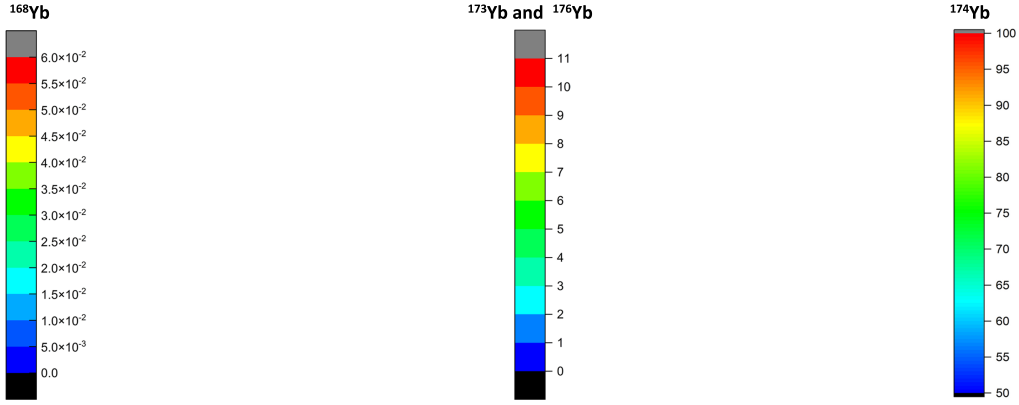


Fig. 6. Degree of enrichment scale for Fig. 5.

isotope mixture will be rather low (∼2%); therefore, their presence does not affect the utility of ^{175}Yb for PRRT.

From Fig. 5, it can be found that, for the full angle divergence of the atomic beam of 60°, when the atomic number densities are restricted 1×10^{11} atoms/cm^3, the degree of enrichment of 94% of ^{174}Yb can be achieved for the lasers with a bandwidth of 500 MHz. Under the same conditions, when the bandwidth of the lasers is reduced to 200 MHz, the degree of enrichment

increases to 97%. When the number density of the atomic ensemble is increased to 5×10^{12} atoms/cm^3, the degree of enrichment drops to 70%. Therefore, in order to achieve the degree of enrichment ~90%, it is necessary to restrict the number density of the atomic ensemble to $\leq 1 \times 10^{12}$ atoms/cm^3. Further, the excitation lasers with a bandwidth up to 500 MHz can be used for the isotope separation process.

D. Production Rates

Production rates can be calculated using the following expression:

$$P\left(\frac{g}{\text{hour}}\right) = 2.827 \times 10^3 \left(b^2.p.l.d.A.f.\eta.i.n.\frac{M}{N_A}.\text{PRF}\right), \tag{10}$$

where b is the laser beam diameter (cm), p is fractional population of level $|1\rangle$, l is the length of the laser–tom interaction region (cm), d is the number density of atoms in the interaction region (atoms/ cm^3), A is the fractional abundance of the target isotope, f is the fractional flux (flux relative to the flux of unhindered atomic beam), η is the ionization efficiency (derived from the density matrix calculations), I is the irradiation probability, n is the number of passes of the laser beam through the atomic ensemble, M is the atomic mass of the target isotope (grams), N_A is the Avogadro number ($6.02214076 \times 10^{23}$), and PRF is the pulse repetition frequency of the lasers (Hz).

For the values of $b = 3$ cm, $p = 1$, $l = 27$ cm, $d = 1 \times 10^{11}$ atoms/cm^3, $A = 0.3183$, $f = 1$, $\eta = 0.5856$, $i = 0.85$, $n = 1$, $M = 174$, PRF = 10 kHz, the production rate was calculated to be 31 mg/h. When the number density of the atomic ensemble is increased to 5×10^{11} atoms/cm^3 and 1×10^{12} atoms/cm^3, the production rate can be increased to 157 mg/h and 314 mg/h, respectively.

The peak power densities of the excitation and ionization lasers to obtain the above production rates are 300, 600, and 50,000 W/cm^2, which correspond to average powers of 0.64 W, 1.27 W, and 106 W, respectively for the 30 ns pulse width, 30 mm diameter, 10 kHz pulse repetition rate lasers. Though the average powers of the excitation lasers can be easily achievable in any laboratory, obtaining an average power of 106 W for the ionization laser is not an easy task. Very few laboratories in the world have achieved such high average powers [29]. Therefore, dependence of ionization efficiency on the peak power density has been studied. For this, ionization efficiency calculations have been carried out varying peak power densities of the ionization laser, and the results are plotted in Fig. 7. From Fig. 7, when the peak power density of the ionization laser is reduced to an achievable value of 5000 W/cm^2 (average power = 10.6 W), the ionization efficiency falls to 0.088. Under this condition, the production rates corresponding to number densities of 1×10^{11}, 5×10^{11}, and 1×10^{12} atoms/cm^3 have been calculated to be 4.8 mg/h, 24 mg/h, and 48 mg/h respectively. The production rate in this case is comparable to the value of 20 mg/h reported previously for the enrichment of ^{176}Yb [19,23].

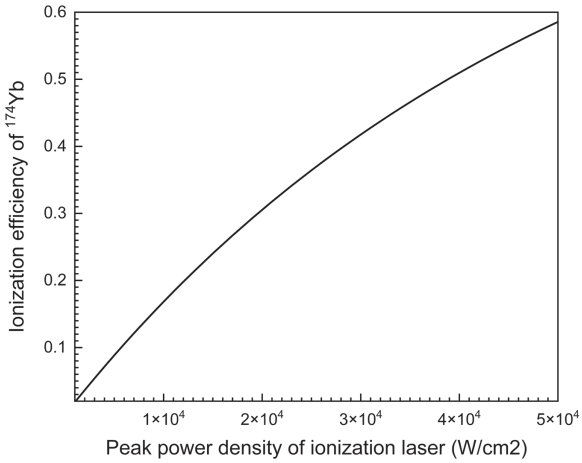


Fig. 7. Dependence of ionization efficiency on the peak power density of the ionization laser. Full angle divergence of the atomic beam is 60°. The peak power densities of the excitation lasers are 300 and 600 W/cm^2, and the bandwidth of both excitation lasers is 500 MHz.

E. Analysis of Irradiated Enriched Isotope Mixture

In this section, we theoretically analyze the suitability of the enriched isotope mixture for the PRRT. The production efficiency of a radionuclide through neutron irradiation in nuclear reactor can be given by the radionuclide production efficiency equation:

$$\varepsilon = \left[\frac{\sigma\varphi\ \left\{1 - e^{-\lambda t_1}\right\}}{\lambda}\right] - e^{(-\lambda t_2)}, \tag{11}$$

where λ is the decay constant, σ is the thermal neutron absorption cross section (cm^2), ϕ is the thermal neutron flux (neutrons/cm^2 − s), t_1 is the time of irradiation, and t_2 is the time of cooling after irradiation. The decay constant λ can be related to the half-life of the radio nuclide using the following expression:

$$\lambda = \frac{\ln 2}{T_{1/2}}, \tag{12}$$

where $T_{1/2}$ is the half-life of the isotope.

For the analysis, we consider the isotope abundances ^{168}Yb (1.2×10^{-2}%), ^{170}Yb (0.29%), ^{171}Yb (1.4%), ^{172}Yb (4.1%), ^{173}Yb (2.9%), ^{174}Yb (88.6%), and ^{176}Yb (2.7%) corresponding to laser isotope separation at the number density of 1×10^{12} atoms/cm^3 and compare them with the 90% enriched ^{176}Yb [19,23] and 68.9% enriched ^{176}Lu [5,6] for the production of radionuclide for PRRT by irradiation in a low flux nuclear reactor. The comparison is tabulated in Table 4. It can seen in Table 4 that the specific activity ^{175}Yb produced is 2 orders lower than the specific activity produced with 68.9% enriched ^{176}Lu and more than 1 order higher than that produced by irradiation of 90% enriched ^{176}Yb. Further, 1 g of 88.6% enriched ^{174}Yb produces 759 doses of PRRT isotope with activity of 7.4 GBq each.

It is also important to note that the enriched ^{174}Yb isotope mixture also consists of other isotopes, viz., ^{168}Yb (1.2×10^{-2}%) and ^{176}Yb (2.7%) in minor quantities, that upon irradiation produce radionuclides such as ^{169}Yb

Table 4. Table of Comparison of Daughter Nuclides Produced from the Enriched Isotope Mixtures of Yb and Lu in Low Flux Nuclear Reactors

Description	^{174}Yb	^{176}Yb	^{176}Lu
Degree of enrichment (%)	88.6%	90.0%	68.9%
Neutron flux of the reactor (neutrons/cm^2/sec)	3×10^{13}	3×10^{13}	3×10^{13}
Irradiation period (days)	21	21	21
Cooling time (days)	0	0	0
Conversion efficiency to daughter nuclide	9.51×10^{-4} (^{175}Yb)	6.29×10^{-5} (^{177}Lu)	4.61×10^{-2} (^{177}Lu)
Amount of radioisotope produced per gram of enriched isotope mixture (gm)	8.42×10^{-4}	5.66×10^{-5}	3.18×10^{-2}
No carrier added specific activity (Bq/gm)	6.6716×10^{15}	4.1037×10^{15}	4.1037×10^{15}
Specific activity of the irradiated isotope mixture (Bq/gm)	5.62×10^{12}	2.32×10^{11}	1.30×10^{14}
Weight for single dose 7.4 GBq activity (grams)	1.31×10^{-3}	31.83×10^{-2}	5.67×10^{-5}
Number of doses (7.4 GBq) produced per 1 g of irradiation	759	31	17,633

($T_{1/2} = 32.02$ days) and ^{177}Lu ($T_{1/2} = 6.65$ days). The calculated radionuclide purity of ^{175}Yb at the end of irradiation was found to be 98.6%. Due to the difference in the half-lives of the radionuclides in the irradiated isotopic mixture (^{169}Yb, ^{175}Y, and ^{177}Lu) radionuclidic purity of ^{175}Yb falls to 96.2% after 7 days of irradiation.

Based on the ^{175}Yb activity produced upon irradiation of the enriched isotope mixture in a low flux nuclear reactor, its radionuclidic purity, and its favorable decay characteristics, this radionuclide can be used for PRRT. Since the precursor ^{174}Yb isotope can be sufficiently enriched using broadband lasers with adequately high production rates and the ability to irradiate in low flux reactors for the production of PRRT drug, it is envisaged that ^{175}Yb could evolve as the most economical and widely available PRRT drug for cancer therapy.

4. CONCLUSION

A three-step photoionization scheme has been studied for the laser isotope separation of ^{174}Yb from natural Yb. It has been shown that it is possible to enrich the ^{174}Yb isotope to ~90% using broadband lasers with a bandwidth of 500 MHz unlike laser isotope separation of ^{176}Lu, ^{177}Lu, and ^{176}Yb, wherein the bandwidth of excitation lasers needs to be restricted to ≤ 100 MHz. It has also been shown that, up to 60° full angle divergence of the atomic ensemble, the Doppler broadening of the atomic ensemble does not impact the degree of enrichment. The optimum peak densities for the excitation and ionization lasers and operable range of the atomic number densities have been derived. It has been further shown that it is possible to produce ~90% enriched ^{174}Yb at a rate up to 48 mg/h. On the whole, the system described for the laser isotope separation of ^{174}Yb is much simpler than that required for the separation of ^{176}Lu, ^{177}Lu, and ^{176}Yb isotopes. The enriched isotope mixture can be irradiated in the widely available low flux reactors for the production of ^{175}Yb radionuclide for PRRT. As a result of the simpler laser isotope separation process and ability to irradiate in low flux reactors, it is envisaged that production of the ^{175}Yb isotope will be economical and widely available to the public at large as a PRRT drug for cancer therapy.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

REFERENCES

1. S. J. Goldsmith, "Targeted radionuclide therapy: a historical and personal review," Sem. Nuclear Med. **50**, 87–97 (2020).
2. R. Zimmermann, *Nuclear Medicine-Radioactivity for Diagnosis and Therapy*, 2nd ed. (EDP Sciences, 2017).
3. A. Dash, M. Raghavan, A. Pillai, and F. F. Knapp, Jr., "Production of ^{177}Lu for targeted radionuclide therapy: available options," Nucl. Med. Mol. Imag. **49**, 85–107 (2015).
4. A. Dash, R. Chakravarthy, F. R. Knapp, Jr., and A. M. R. Pillai, "Indirect production of No carrier added (NCA) ^{177}Lu from irradiation of enriched ^{176}Yb-options for ytterbium/lutetium separation," Curr. Radiopharm. **8**, 107–118 (2015).
5. A. B. D'yachkov, S. K. Kovalevich, A. V. Labozin, V. P. Labozin, S. M. Mironov, V. Y. Panchenko, V. A. Firsov, G. O. Tsvetkov, and G. G. Shatalova, "Selective photoionisation of lutetium isotopes," Quantum Electron. **42**, 953–956 (2012).
6. M. V. Suryanarayana, "Isotope selective three-step photoionization of ^{176}Lu," J. Opt. Soc. Am. **38B**, 353–370 (2021).
7. A. B. D'yachkov, V. A. Firsov, A. A. Gorkunov, A. V. Labozin, S. M. Mironov, V. Y. Panchenko, A. N. Semenov, G. G. Shatalovaa, and G. O. Tsvetkov, "Photoionization spectroscopy for laser extraction of the radioactive isotope ^{177}Lu," Appl. Phys. B **121**, 425–431 (2015).
8. A. B. D'yachkov, A. A. Gorkunov, A. V. Labozin, S. M. Mironov, V. Y. Panchenko, V. A. Firsov, and G. O. Tsvetkov, "Effect of amplified spontaneous emission on selectivity of laser photoionisation of the ^{177}Lu radioisotope," Quantum Electron. **46**, 574–577 (2016).
9. A. B. D'yachkov, A. A. Gorkunov, A. V. Labozin, S. M. Mironov, G. O. Tsvetkov, V. Y. Panchenko, and V. A. Firsov, "Study of a selective photoionization scheme of ^{177}Lu," Opt. Spectrosc. **125**, 839–844 (2018).
10. I. V. Ageeva, A. B. D'yachkov, A. A. Gorkunov, A. V. Labozin, S. M. Mironov, V. Y. Panchenko, V. A. Firsov, G. O. Tsvetkov, and E. G. Tsvetkov, "Laser photoionisation selectivity of ^{177}Lu radionuclide for medical applications," Quantum Electron. **49**, 832–838 (2019).
11. M. V. Suryanarayana and M. Sankari, "Isotope selective three-step photoionization of ^{177}Lu," J. Opt. Soc. Am. **39B**, 2502–2521 (2022).
12. T. Stigbrand, J. Carlsson, and G. P. Adams, *Targeted Radionuclide Tumor Therapy-Biological Aspects* (Springer, 2008).
13. S. Chakraborty, P. R. Unni, M. Venkatesh, and M. R. A. Pillai, "Feasibility study for production of ^{175}Yb: a promising therapeutic radionuclide," Appl. Radiat. Isot. **57**, 295–301 (2002).
14. R. Chakravarty, T. Das, M. Venkatesh, and A. Dash, "An electro-amalgamation approach to produce ^{175}Yb suitable for

radiopharmaceutical applications," Radiochim. Acta **100**, 255–261 (2012).
15. M. T. Duran, F. Juget, Y. Nedjadi, F. Bochud, Z. Talip, N. P. van der Meulen, U. Koster, C. Duchemin, T. Stora, and C. Bailat, "Ytterbium-175 half-life determination," Appl. Radiat. Isot. **176**, 109893 (2021).
16. S. K. Borisov, M. A. Kuzmina, and V. A. Mishin, "A study of isotopically selective photoionization of ytterbium atoms for laser isotope separation," J. Russ. Laser Res. **17**, 332–345 (1996).
17. V. I. Derzhiev, V. A. Kuznetsov, L. A. Mikhal'tsov, V. M. Mushta, A. Y. Sapozhkov, A. N. Tkachev, S. A. Chaushanskii, and S. I. Yakovlenko, "Laser separation of highly enriched ^{168}Yb in weighable quantities," Quantum Electron. **26**, 751–752 (1996).
18. M. Sankari and M. V. Suryanarayana, "Studies on the isotope selective photoionization of the low-abundant ^{168}Yb isotope," J. Phys. B **31**, 261–273 (1998).
19. O. I. Andreev, V. I. Derzhiev, V. M. Dyakin, A. G. Egorov, L. A. Mikhal'tsov, V. A. Tarasov, A. I. Tolkachev, Y. G. Toporov, S. A. Chaushanskii, and S. I. Yakovlenko, "Production of a highly enriched Yb isotope in weight amounts by the atomic-vapor laser isotope separation method," Quantum Electron. **36**, 84–89 (2006).
20. H. Park, D.-H. Kwon, Y. Cha, S. Nam, T.-S. Kim, J. Han, Y. Rhee, D.-Y. Jeong, and C.-J. Kim, "Laser isotope separation of ^{176}Yb for medical applications," J. Korean Phys. Soc. **49**, 382–386 (2006).
21. H. Park, D.-H. Kwon, Y. H. Cha, T.-S. Kim, J. Han, K.-H. Ko, D.-Y. Jeong, and C.-J. Kim, "Stable isotope production of ^{168}Yb and ^{176}Yb for industrial and medical applications," J. Nucl. Sci. Technol. **45**, 111–116 (2008).
22. M. Sankari, P. V. Kiran Kumar, and M. V. Suryanarayana, "Study of line shapes in the selective ionization of Yb^{176} isotope in a two-step resonance, three-step ionization scheme," J. Opt. Soc. Am. B **25**, 1820–1828 (2008).
23. M. V. Suryanarayana and M. Sankari, "Isotope selective three-step photoionization of ^{176}Yb," J. Opt. Soc. Am. **38B**, 3331–3339 (2021).
24. B. W. Shore, *The Theory of Coherent Atomic Excitation: Simple Atoms and Fields* (Wiley, 1990), Vol. **1**.
25. P. Lambropoulos and A. Lyras, "Theory of resonant ionization by broad-band radiation in the determination of isotopic abundances," Phys. Rev. A **40**, 2199–2202 (1989).
26. A. Lyras, B. Zorman, and P. Lambropoulos, "Theory of doubly resonant ionization by broad-band radiation applied to the determination of isotopic abundances," Phys. Rev. A **42**, 543–549 (1990).
27. P. Kusch and V. W. Hughes, "Atomic and molecular beam spectroscopy," in *Handbuch der Physik*, S. Flugge, ed. (Springer-Verlag, 1959).
28. T. Shibata, K. Tamura, and K. Ogura, "Symmetric charge transfer cross section of ytterbium, lutetium and samarium," Japan atomic energy agency report 076-2008 (2008).
29. A. B. D'yachkov, A. A. Gorkunov, A. V. Labozin, S. M. Mironov, V. Y. Panchenko, V. A. Firsov, and G. O. Tsvetkov, "Development of a laser system of the laboratory AVLIS complex for producing isotopes and radionuclides," Quantum Electron. **48**, 75–81 (2018).